

# **ASSIGNMENT**

## **Subject: GUI Programming Using C# .Net**

### **Long Questions**

#### **UNIT-I**

1. What is .Net Technology? Explain the benefits of .NET.
2. Explain the architecture of the .NET Framework with the diagram.
3. Explain CLR and its architecture in detail.
4. Differentiate between Managed Code and Unmanaged Code. Provide examples.
5. What is JIT Compiler? Explain its types and compare JIT Compiler with Traditional Compiler.
6. Explain CTS (Common Type System) with Value Types and Reference Types.
7. What is CLS (Common Language Specification)? Explain its role in the .NET Framework.
8. Explain FCL (Framework Class Library), MSIL and their role in the .NET Framework.
9. Describe the role of Assemblies and Metadata in the .NET Framework.
10. Explain the compilation process of C# source code into executable format in .NET.
11. Compare .NET technology with other current technologies like Java, PHP, Python, and Node.js.

#### **UNIT-II**

1. Explain the concept of Object-Oriented Programming in C# with its advantages.
2. What are delegates in C#? Explain single-cast and multicast delegates with examples.
3. What are arrays in C#? Explain jagged arrays with examples.
4. Define indexers and properties in C# and explain with examples.
5. Explain ArrayList and Hashtable including their methods with examples.
6. Explain the concept of Constructor and Destructor in C# with examples.
7. Explain Types of inheritance in C# with example.
8. Explain method overloading and method overriding with suitable example.
9. Explain exception handling in C# using example.
10. Explain process to call a base class constructor from a derived class with example.
11. Explain the concept of abstraction in C# using class and interface.
12. What is a string in C#? Explain common string handling methods with syntax and output.
13. What are the different data types in C#? Explain each with suitable examples.
14. Describe all types of operators in C# with examples.
15. Explain the different categories of Toolbox controls available in Visual Studio.
16. Explain Design Window, Solution Explorer, Server Explorer, Property and Event Explorer, Class View and Command Window.
17. Explain the menus and important functionalities of the Visual Studio Tool.

#### **UNIT-III**

1. Explain 8 common controls in detail with their usage, key properties, methods, and events.
2. Explain different types of dialog box controls in detail.
3. Describe the MessageBox class with all its available buttons, icons, and usage examples.
4. Explain the RichTextBox control with its important properties, methods, and events.
5. Explain MDI Forms in .NET with their advantages and working.
6. Explain Menu Bar and Context Menu in Windows Forms with suitable examples.

7. **Explain Properties, Methods and Events of Windows Forms controls in detail.**
8. **Explain the Timer control with a suitable example.**
9. Explain the List View and Tree View controls along with its important properties, methods, and events.
10. Explain the GroupBox and Panel container controls with their key properties.
11. Describe the ProgressBar, control along with its important properties, methods and events.
12. Explain the DateTimePicker control and MonthCalendar with properties, methods, and events.

#### **UNIT-IV**

1. **Explain the architecture of ADO.NET.**
2. **Explain .NET Data Provider and its components in detail.**
3. **Explain the CRUD operations using SqlCommand objects with code examples.**
4. **Explain use of DataAdapter and DataSet with database using examples.**
5. **Explain DataReader in ADO.NET with suitable examples.**
6. **Explain ExecuteReader, ExecuteNonQuery and ExecuteScalar with suitable examples.**
7. **Discuss the DataGridView Control in detail with its properties, methods, and events.**
8. **What are Crystal Reports? Explain the different sections of Crystal Report in detail.**
9. Explain Connected and Disconnected data access models in ADO.NET with examples.
10. Explain the steps to establish a connection with SQL Server in C#.
11. Explain designing, formatting, grouping, parameters and summarizing data in Crystal Reports.
12. Discuss the Setup and Deployment process of a .NET application in Visual Studio.

### **Short Questions**

#### **UNIT-I**

1. What is .NET?
2. Who developed the .NET Framework?
3. Name any two programming languages supported by .NET.
4. What is the CLR?
5. What is MSIL or CIL?
6. What is the purpose of JIT Compiler?
7. What is the difference between JIT and Traditional compiler?
8. What is the role of CTS in .NET?
9. What is CLS?
10. What do you mean by Managed Code?
11. What is Unmanaged Code?
12. What is a Namespace?
13. Define an Assembly.
14. What is Metadata in .NET?
15. What is the purpose of the Garbage Collector?
16. Name any three types of applications that can be developed using .NET.
17. What is Visual Studio used for?
18. Name two important namespaces in .NET.
19. What is the purpose of ADO.NET?
20. What is the use of COM Interop?

## **UNIT-II**

1. What is the use of Solution Explorer in Visual Studio?
2. Differentiate between Code Window and Design Window.
3. What is the purpose of the Toolbox in a Windows Forms application?
4. What is IntelliSense in Visual Studio?
5. What is the function of the Properties Window?
6. Define a class and an object in C#.
7. What is the difference between a field and a property?
8. List any four built-in value types in C#.
9. What is an auto-implemented property? Give an example.
10. What is the use of a constructor in C#?
11. What is the difference between static and instance methods?
12. What is a Jagged Array? How is it different from a 2D array?
13. What is the purpose of the finally block in exception handling?
14. Write the syntax to declare a delegate in C#.
15. What is a multicast delegate?
16. What is the role of the throw keyword in C#?
17. How is a string different from a StringBuilder?
18. What is the difference between ArrayList and Hashtable?
19. What is the output of "hello".ToUpper()?
20. What is an indexer in C# and how is it accessed?
21. What is the syntax to inherit a class in C#?
22. Define method overloading with an example.
23. What keyword is used to allow a method to be overridden in a derived class?
24. What does the base keyword do in a constructor?
25. How does method overriding differ from method hiding (new)?
26. Can constructors be overloaded in C#?
27. What is Interface?
28. What is Abstract class?
29. Method Hiding.

## **UNIT-III**

### **DIFFERENCES**

- Diff: ListBox vs ComboBox
  - Diff: CheckBox vs RadioButton
  - Diff: ListBox vs CheckedListBox
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1. What is a ListBox?
  2. Write the syntax to add an item to a ComboBox.
  3. What does the Checked property of a CheckBox do?
  4. What is the use of the SelectedItem property in ListBox?
  5. Which event is fired when the selected index of a ComboBox changes?
  6. What is the purpose of a RichTextBox?
  7. Name any two properties of a TextBox.
  8. What is the default view of a ListView control?
  9. How can you expand all nodes in a TreeView?
  10. What does the ImageLocation property of PictureBox do?
  11. Write the syntax to show a message using MessageBox.
  12. What is the use of DateTimePicker control?
  13. What is the difference between Checked and Selected properties?

14. What does ShowDialog() do in OpenFileDialog?
15. What is the use of FontDialog control?
16. Define the ToolTip control.
17. What does ErrorProvider do?
18. What is the function of NotifyIcon control?
19. How do you navigate to a URL using WebBrowser?
20. What is the use of ProgressBar control?
21. What is the role of the Timer control?
22. Which property of Timer sets the interval?
23. Write the syntax to add a node to a TreeView.
24. What is the use of Panel control?
25. What is the ContextMenuStrip used for?
26. What is the TabControl control used for?
27. Define the Items collection in ComboBox.
28. How do you clear all items in a ListBox?
29. What does the Value property of DateTimePicker return?
30. Which dialog is used to preview a document before printing?

#### **UNIT-IV**

1. Define ADO.NET.
2. What is the role of **SqlConnection** class?
3. Differentiate between **SqlCommand** and **SqlDataAdapter**.
4. What is a **DataSet**? How is it different from a **DataTable**?
5. List any four important **properties of DataGridView control**.
6. Define **Data Binding**.
7. What is a **Connection String**? Give an example.
8. What is the default file extension of a Crystal Report?
9. Name any two **advantages of Crystal Reports**.
10. What is a **DataReader**? How is it different from a DataAdapter?
11. Expand **CRUD** and explain its significance.
12. What is the function of **App.config** file?
13. Name two **types of deployment** in .NET.
14. List two **advantages of disconnected architecture**.
15. What is **parameterized query** in SqlCommand?
16. List the **Crystal Reports sections**.
17. Which class in ADO.NET is used to fill a DataSet?
18. What is **Exporting Reports** in Crystal Reports?
19. Define **Transaction** in ADO.NET.